

Summer School Lecture Notes: Semidefinite Programming in Quantum Information

Lecture 2: Applications — PPT Relaxations and the Diamond Norm

Reading (inspiration): These notes follow standard conventions in quantum information and semidefinite programming; they are inspired by lecture notes of John Watrous.

1 Scope and goals of Lecture 2

Lecture 1 developed the “calculus” of SDPs (standard forms, Lagrangians, duality, Slater, complementary slackness). In Lecture 2 we apply this calculus to two canonical quantum information problems in which SDPs serve as *relaxations* or *exact* characterizations:

- (i) optimizing over separable states via the *PPT relaxation* (partial transposition),
- (ii) computing the distinguishability of channels via the *diamond norm* (completely bounded trace norm).

The first example illustrates a central concept in modern QIT (and optimization in general!): SDPs are a powerful way of relaxing hard computational problems and “convexifying them” without losing much (and of often provably so!). The second example illustrates another fact: problems that are not obviously SDPs at first sight can be mapped to them through clever matrix analysis tricks. Both examples illustrate a theme that will repeat throughout the lectures: *the primal is an optimization over a “physical” set, while the dual is a witness/certificate.*

2 SDPs as relaxations of complex feasible sets

In many quantum information tasks the natural feasible set is convex but has an implicit or complicated description, the prototypical example being the set of separable states. A typical pattern is that we wish to optimize some expectation value over such a complicated set, e.g.

$$\max\{\mathrm{Tr}(H\rho) : \rho \in \mathcal{K}\},$$

where H is a fixed Hermitian operator and $\mathcal{K}$ is a convex set of states, measurements, or channels.

Relaxation viewpoint. If $\mathcal{K} \subseteq \mathcal{K}_{\text{relax}}$ then

$$\max_{\rho \in \mathcal{K}} \mathrm{Tr}(H\rho) \leq \max_{\rho \in \mathcal{K}_{\text{relax}}} \mathrm{Tr}(H\rho),$$

so solving the relaxed problem gives an *upper bound* on the true optimum. If the relaxed set is defined by semidefinite constraints, we obtain an efficiently computable bound *plus a dual certificate*.

Why duals matter. If you can exhibit a dual feasible point with objective value u , you have proved that *no* feasible object in $\mathcal{K}_{\text{relax}}$ can exceed u . When you are lucky, complementary slackness can also tell you *what* optimal solutions must look like.

3 Bipartite entanglement and the PPT relaxation

Let us dive deeper into this idea in the case of separable states.

3.1 Separable states and partial transposition

Let $\mathcal{X}$ and $\mathcal{Y}$ be finite-dimensional complex Hilbert spaces, and consider bipartite states on $\mathcal{X} \otimes \mathcal{Y}$.

Definition 3.1 (Separable states). A density operator $\rho \in \mathcal{D}(\mathcal{X} \otimes \mathcal{Y})$ is *separable* if it can be written as a convex combination of product states:

$$\rho = \sum_k p_k \sigma_k \otimes \tau_k, \quad p_k \geq 0, \quad \sum_k p_k = 1, \quad \sigma_k \in \mathcal{D}(\mathcal{X}), \quad \tau_k \in \mathcal{D}(\mathcal{Y}).$$

We denote the set of separable PSD operators by $\text{Sep}(\mathcal{X} : \mathcal{Y}) \subseteq \text{Pos}(\mathcal{X} \otimes \mathcal{Y})$ and the set of separable states by $\text{Sep}(\mathcal{X} : \mathcal{Y}) \cap \mathcal{D}(\mathcal{X} \otimes \mathcal{Y})$.

The set $\text{Sep}(\mathcal{X} : \mathcal{Y})$ is convex and closed, but its exact characterization is hard beyond low dimensions.

Definition 3.2 (Transpose and partial transpose). Fix orthonormal bases for $\mathcal{X}$ and $\mathcal{Y}$. The transpose map $T : \mathcal{L}(\mathcal{X}) \rightarrow \mathcal{L}(\mathcal{X})$ is defined by $T(X) = X^\top$. The *partial transpose on $\mathcal{X}$* is the linear map

$$T \otimes \text{id}_{\mathcal{L}(\mathcal{Y})} : \mathcal{L}(\mathcal{X} \otimes \mathcal{Y}) \rightarrow \mathcal{L}(\mathcal{X} \otimes \mathcal{Y}).$$

(One similarly defines partial transpose on $\mathcal{Y}$.)

Definition 3.3 (PPT operators). An operator $P \in \text{Pos}(\mathcal{X} \otimes \mathcal{Y})$ is *PPT* (has *positive partial transpose*) if

$$(T \otimes \text{id}_{\mathcal{L}(\mathcal{Y})})(P) \succeq 0.$$

We write

$$\text{PPT}(\mathcal{X} : \mathcal{Y}) = \{ P \in \text{Pos}(\mathcal{X} \otimes \mathcal{Y}) : (T \otimes \text{id})(P) \succeq 0 \}.$$

Proposition 3.4 (Peres criterion: separable $\Rightarrow$ PPT). *Every separable PSD operator is PPT:*

$$\text{Sep}(\mathcal{X} : \mathcal{Y}) \subseteq \text{PPT}(\mathcal{X} : \mathcal{Y}).$$

Proof. If $P = \sum_k p_k \sigma_k \otimes \tau_k$ is separable and PSD, then

$$(T \otimes \text{id})(P) = \sum_k p_k \sigma_k^\top \otimes \tau_k.$$

Because the transpose map is positive, $\sigma_k^\top \succeq 0$ whenever $\sigma_k \succeq 0$, and hence each term in the sum is PSD, so the sum is PSD. $\square$

Remark 3.5. The inclusion is strict in general: there exist PPT states that are nevertheless entangled (“bound entanglement”). Thus PPT is a *necessary but not sufficient* condition for separability. It is only sufficient for very low dimensions.

3.2 Optimizing over PPT states as an SDP

Let $H \in \text{Herm}(\mathcal{X} \otimes \mathcal{Y})$. Define the *separable value* and its *PPT relaxation*:

$$\omega_{\text{sep}}(H) := \max\{ \text{Tr}(H\rho) : \rho \in \text{D}(\mathcal{X} \otimes \mathcal{Y}) \cap \text{Sep}(\mathcal{X} : \mathcal{Y}) \},$$

$$\omega_{\text{ppt}}(H) := \max\{ \text{Tr}(H\rho) : \rho \in \text{D}(\mathcal{X} \otimes \mathcal{Y}) \cap \text{PPT}(\mathcal{X} : \mathcal{Y}) \}.$$

By $\text{Sep} \subseteq \text{PPT}$ we have $\omega_{\text{sep}}(H) \leq \omega_{\text{ppt}}(H)$.

Primal SDP (PPT relaxation). The quantity $\omega_{\text{ppt}}(H)$ is the optimal value of the SDP

$$\begin{aligned} & \text{maximize: } \text{Tr}(H\rho) \\ & \text{subject to: } \text{Tr}(\rho) = 1, \\ & \quad \rho \succeq 0, \\ & \quad (\text{T} \otimes \text{id})(\rho) \succeq 0. \end{aligned}$$

This is an SDP because $(\text{T} \otimes \text{id})$ is a fixed linear map.

Dual SDP and decomposable witnesses. Let us derive a clean dual form, because it illustrates how witness structure emerges automatically. Introduce:

- a scalar multiplier $\lambda \in \mathbb{R}$ for $\text{Tr}(\rho) = 1$,
- a slack $P \succeq 0$ for $\rho \succeq 0$,
- a slack $Q \succeq 0$ for $(\text{T} \otimes \text{id})(\rho) \succeq 0$.

The Lagrangian becomes

$$\mathcal{L}(\rho, \lambda, P, Q) = \text{Tr}(H\rho) + \lambda(1 - \text{Tr}(\rho)) - \text{Tr}(P\rho) - \text{Tr}(Q(\text{T} \otimes \text{id})(\rho)).$$

Using the adjoint relation $\text{Tr}(Q(\text{T} \otimes \text{id})(\rho)) = \text{Tr}((\text{T} \otimes \text{id})(Q)\rho)$ (transpose is self-adjoint), we obtain

$$\mathcal{L}(\rho, \lambda, P, Q) = \lambda + \text{Tr}\left((H - \lambda\mathbb{1} - P - (\text{T} \otimes \text{id})(Q))\rho\right).$$

Taking $\sup_{\rho}$ forces

$$\lambda\mathbb{1} - H = P + (\text{T} \otimes \text{id})(Q) \quad \text{for some } P, Q \succeq 0.$$

Therefore the dual may be written as

$$\begin{aligned} & \text{minimize: } \lambda \\ & \text{subject to: } \lambda\mathbb{1} - H = P + (\text{T} \otimes \text{id})(Q), \\ & \quad P \succeq 0, \quad Q \succeq 0, \quad \lambda \in \mathbb{R}. \end{aligned}$$

Equivalently, setting $W := \lambda\mathbb{1} - H$, the dual searches for a decomposition

$$W = P + (\text{T} \otimes \text{id})(Q), \quad P, Q \succeq 0,$$

which is exactly the form of a *decomposable entanglement witness*. Any feasible (λ, P, Q) certifies the upper bound $\omega_{\text{ppt}}(H) \leq \lambda$ by weak duality.

Remark 3.6 (Feasibility and strong duality). Strict feasibility is typically easy: take $\rho_0 = \mathbb{1}/(\dim(\mathcal{X}) \dim(\mathcal{Y}))$, which is full-rank and also PPT. Hence Slater's condition holds for most PPT optimization problems of interest, implying no duality gap.

3.3 Example: overlap with a maximally entangled state

Let $\mathcal{X} = \mathcal{Y} = \mathbb{C}^n$ and let $\tau \in \mathcal{D}(\mathcal{X} \otimes \mathcal{Y})$ be the maximally entangled state

$$\tau = \frac{1}{n} \text{vec}(\mathbf{1}) \text{vec}(\mathbf{1})^*.$$

Consider the objective $H = \tau$ (“singlet fraction”). The PPT relaxation yields an absolute bound:

$$\max\{\text{Tr}(\tau\rho) : \rho \in \mathcal{D}(\mathcal{X} \otimes \mathcal{Y}) \cap \text{PPT}(\mathcal{X} : \mathcal{Y})\} \leq \frac{1}{n}.$$

One quick proof uses the dual perspective: τ has partial transpose proportional to the swap operator, whose operator norm is $1/n$, and PPT implies positivity after partial transpose.

Remark 3.7. This is representative of a broader phenomenon: PPT relaxations often reproduce the “right” scaling in dimension even when they are not exact.

4 The diamond norm and SDPs for channel distinguishability

We will now show how to formulate the diamond norm, which quantifies the optimal distinguishability between two channels, as an SDP. This will be a prime example of a quantity that is not obviously an SDP at first, but becomes one after various smart tricks.

4.1 Norms for linear maps

Let $\Phi : \mathcal{L}(\mathcal{X}) \rightarrow \mathcal{L}(\mathcal{Y})$ be linear. Recall the trace norm on operators: $\|X\|_1 = \text{Tr}(\sqrt{X^*X})$. A naive way to define a norm for maps is the induced trace norm

$$\|\Phi\|_{1 \rightarrow 1} := \max\{\|\Phi(X)\|_1 : \|X\|_1 \leq 1\}.$$

This norm is *not* stable under tensoring with an identity map: $\|\Phi \otimes \text{id}\|_{1 \rightarrow 1}$ can be strictly larger than $\|\Phi\|_{1 \rightarrow 1}$.

Definition 4.1 (Completely bounded trace norm / diamond norm). The *completely bounded trace norm* of Φ is

$$\|\Phi\|_{\diamond} := \sup_{k \geq 1} \|\Phi \otimes \text{id}_{\mathbb{C}^k}\|_{1 \rightarrow 1}.$$

When Φ is Hermiticity-preserving, $\|\Phi\|_{\diamond}$ is often called the *diamond norm* and denoted $\|\Phi\|_{\diamond}$.

Remark 4.2. For maps $\Phi : \mathcal{L}(\mathcal{X}) \rightarrow \mathcal{L}(\mathcal{Y})$ it is enough to take $k = \dim(\mathcal{X})$ in the supremum. In practice, $\|\Phi\|_{\diamond}$ can be computed efficiently via semidefinite programming.

4.2 Operational meaning: channel discrimination

Suppose we are promised that an unknown channel is either Φ_0 or Φ_1 , each with prior probability $1/2$. We may prepare an input state on $\mathcal{X} \otimes \mathcal{Z}$, send $\mathcal{X}$ through the channel, and perform an optimal measurement on $\mathcal{Y} \otimes \mathcal{Z}$.

Theorem 4.3 (Channel analogue of the Holevo–Helström theorem). *For channels $\Phi_0, \Phi_1 : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$ with equal priors, the optimal success probability is*

$$p_{\text{succ}}^* = \frac{1}{2} + \frac{1}{4} \left\| \Phi_0 - \Phi_1 \right\|_1.$$

Thus, computing the diamond norm of the difference of two channels exactly quantifies their distinguishability (with entangled inputs allowed).

4.3 From Stinespring representations to an SDP

The key to turning completely bounded norms into SDPs is to (i) represent the map in a form that exposes a *single* ancillary space, and then (ii) reduce the relevant trace-norm optimization to a *fidelity* maximization, which itself admits a clean SDP.

4.3.1 Stinespring representations for completely positive maps

Recall that a map $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$ is completely positive (CP) if it admits a Kraus representation

$$\Phi(X) = \sum_{a=1}^r K_a X K_a^*,$$

for some $K_a \in L(\mathcal{X}, \mathcal{Y})$. If Φ is trace-preserving, then $\sum_a K_a^* K_a = \mathbb{1}_{\mathcal{X}}$.

A convenient way to package the Kraus operators is as a single linear operator $V : \mathcal{X} \rightarrow \mathcal{Y} \otimes \mathcal{Z}$, where $\mathcal{Z} \cong \mathbb{C}^r$ has orthonormal basis $\{|a\rangle\}_{a=1}^r$ and

$$V = \sum_{a=1}^r K_a \otimes |a\rangle.$$

Then one checks immediately that

$$\Phi(X) = \text{Tr}_{\mathcal{Z}}(V X V^*).$$

This is the (finite-dimensional) *Stinespring representation*. When Φ is trace-preserving, V is an isometry ($V^* V = \mathbb{1}_{\mathcal{X}}$).

Remark 4.4 (How big must the environment be?). One may always take $\dim(\mathcal{Z}) \leq \dim(\mathcal{X}) \dim(\mathcal{Y})$ by choosing a Kraus representation with at most $\dim(\mathcal{X}) \dim(\mathcal{Y})$ operators (e.g. from the Choi matrix rank). The minimal $\dim(\mathcal{Z})$ equals the Choi rank of Φ .

4.3.2 A Stinespring pair for an arbitrary linear map

For a *general* linear map $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$ (not necessarily CP), there is an analogous “two-sided” Stinespring representation: there exist a finite-dimensional ancillary space $\mathcal{Z}$ and operators $A_0, A_1 \in L(\mathcal{X}, \mathcal{Y} \otimes \mathcal{Z})$ such that

$$\Phi(X) = \text{Tr}_{\mathcal{Z}}(A_0 X A_1^*) \quad (X \in L(\mathcal{X})). \quad (1)$$

One simple way to see this is to start from an operator-sum representation

$$\Phi(X) = \sum_{a=1}^r A_a X B_a^*,$$

which always exists (e.g. expand Φ on matrix units, or equivalently expand its Choi operator in an operator basis). Setting $\mathcal{Z} \cong \mathbb{C}^r$ with basis $\{|a\rangle\}$ and defining

$$A_0 = \sum_{a=1}^r A_a \otimes |a\rangle, \quad A_1 = \sum_{a=1}^r B_a \otimes |a\rangle,$$

immediately gives (1) after taking the partial trace over $\mathcal{Z}$.

Remark 4.5 (Special cases). (i) If Φ is CP, we may take $A_0 = A_1 = V$ and recover the usual Stinespring form.

(ii) If Φ is Hermiticity-preserving, one can choose a representation with $A_1 = A_0$ up to a simple linear transformation (intuitively, the “left” and “right” Kraus families can be chosen to be adjoints).

Given a Stinespring pair (A_0, A_1) for Φ , define the completely positive maps $\Psi_0, \Psi_1 : \mathcal{L}(\mathcal{X}) \rightarrow \mathcal{L}(\mathcal{Z})$ by

$$\Psi_0(X) = \text{Tr}_{\mathcal{Y}}(A_0 X A_0^*), \quad \Psi_1(X) = \text{Tr}_{\mathcal{Y}}(A_1 X A_1^*). \quad (2)$$

These maps will control the completely bounded trace norm of Φ via a maximum-output *fidelity* optimization.

4.4 An SDP for the fidelity

The fidelity is the bridge between trace norms and semidefinite constraints.

Definition 4.6 (Fidelity). For $P, Q \succeq 0$ acting on the same space, define the (*root*) *fidelity*

$$F(P, Q) := \|\sqrt{P} \sqrt{Q}\|_1.$$

When P and Q are density operators, this equals $\text{Tr} \sqrt{\sqrt{P} Q \sqrt{P}}$. (Some authors define $F(P, Q)$ to be the *square* of this quantity; we will not.)

A useful characterization of positive semidefinite block matrices is the following.

Lemma 4.7 (Block-positivity $\Leftrightarrow$ contraction factorization). *Let $P, Q \succeq 0$ and let X be any operator of compatible dimension. Then*

$$\begin{pmatrix} P & X \\ X^* & Q \end{pmatrix} \succeq 0 \iff \exists K \text{ with } \|K\|_\infty \leq 1 \text{ and } X = \sqrt{P} K \sqrt{Q}.$$

Proof sketch. For the forward direction, apply the Schur complement (first for $P \succ 0$, then by continuity) to obtain $X^* P^{-1} X \preceq Q$ and hence $P^{-1/2} X Q^{-1/2}$ is a contraction. The reverse direction is a direct multiplication: if $X = \sqrt{P} K \sqrt{Q}$ with $\|K\|_\infty \leq 1$, then

$$\begin{pmatrix} \sqrt{P} & 0 \\ 0 & \sqrt{Q} \end{pmatrix} \begin{pmatrix} \mathbb{1} & K \\ K^* & \mathbb{1} \end{pmatrix} \begin{pmatrix} \sqrt{P} & 0 \\ 0 & \sqrt{Q} \end{pmatrix} \succeq 0,$$

because $\begin{pmatrix} \mathbb{1} & K \\ K^* & \mathbb{1} \end{pmatrix} \succeq 0$ holds iff $\|K\|_\infty \leq 1$. □

This lemma immediately yields an SDP for the fidelity.

Theorem 4.8 (SDP for the fidelity). *Let $P, Q \succeq 0$. The fidelity $F(P, Q) = \|\sqrt{P}\sqrt{Q}\|_1$ equals the optimum of the following SDP:*

Primal.

$$\begin{aligned} & \text{maximize: } \frac{1}{2} \text{Tr}(X + X^*) \\ & \text{subject to: } \begin{pmatrix} P & X \\ X^* & Q \end{pmatrix} \succeq 0, \quad X \in \mathcal{L}(\mathcal{H}). \end{aligned}$$

Dual.

$$\begin{aligned} & \text{minimize: } \frac{1}{2} \text{Tr}(PY) + \frac{1}{2} \text{Tr}(QZ) \\ & \text{subject to: } \begin{pmatrix} Y & -\mathbb{1} \\ -\mathbb{1} & Z \end{pmatrix} \succeq 0, \quad Y, Z \in \text{Pos}(\mathcal{H}). \end{aligned}$$

Proof sketch. By Lemma 4.7, primal feasibility is equivalent to $X = \sqrt{P}K\sqrt{Q}$ for a contraction K . The objective becomes $\Re \text{Tr}(\sqrt{P}\sqrt{Q}K)$, whose maximum over contractions equals $\|\sqrt{P}\sqrt{Q}\|_1$ by the variational characterization of the trace norm $\|M\|_1 = \max_{\|U\|_\infty \leq 1} \Re \text{Tr}(U^*M)$. A polar decomposition of $\sqrt{P}\sqrt{Q}$ shows the maximum is attained with K unitary on the support. Deriving the dual follows from the standard Lagrangian method as in Lecture 1. $\square$

Remark 4.9 (Why this SDP is so useful). The constraint is a *single* PSD block matrix, so it composes extremely well with other constraints (such as $\rho \in \mathcal{D}(\mathcal{X})$ or linear constraints defining a relaxation). This is why the fidelity appears frequently in SDP formulations of quantum information tasks.

4.5 Maximum output fidelity and the completely bounded trace norm

Return to a linear map $\Phi : \mathcal{L}(\mathcal{X}) \rightarrow \mathcal{L}(\mathcal{Y})$ and fix a Stinespring pair A_0, A_1 as in (1), with associated CP maps Ψ_0, Ψ_1 defined by (2). We will now state without proof the following characterization of the diamond norm in terms of the fidelity. We refer to [?] for a proof.

Theorem 4.10 (CB trace norm as a maximum output fidelity). *With notation as above,*

$$\|\Phi\|_\diamond = \max \left\{ F(\Psi_0(\rho_0), \Psi_1(\rho_1)) : \rho_0, \rho_1 \in \mathcal{D}(\mathcal{X}) \right\}.$$

If Φ is Hermiticity-preserving, then $\|\Phi\|_\diamond = \|\Phi\|_\diamond$.

Combining Theorem 4.10 with Theorem 4.8 gives a clean SDP for $\|\Phi\|_\diamond$.

Primal SDP (maximum output fidelity).

$$\begin{aligned} & \text{maximize: } \frac{1}{2} \text{Tr}(Y + Y^*) \\ & \text{subject to: } \begin{pmatrix} \Psi_0(\rho_0) & Y \\ Y^* & \Psi_1(\rho_1) \end{pmatrix} \succeq 0, \\ & \quad \rho_0 \in \mathcal{D}(\mathcal{X}), \rho_1 \in \mathcal{D}(\mathcal{X}), Y \in \mathcal{L}(\mathcal{Z}). \end{aligned}$$

Dual SDP (a convenient simplified form). A particularly useful equivalent dual form is

$$\text{minimize: } \frac{1}{2} \left(\|\Psi_0^*(Z)\|_\infty + \|\Psi_1^*(Z^{-1})\|_\infty \right) \quad \text{subject to: } Z \succ 0,$$

where $\|\cdot\|_\infty$ denotes the operator norm and Ψ_i^* is the Hilbert–Schmidt adjoint.

Remark 4.11 (How you use these SDPs). • Any *primal* feasible point gives a *lower bound* on $\|\Phi\|_\diamond$ (a concrete distinguishability strategy).

- Any *dual* feasible point gives an *upper bound* on $\|\Phi\|_\diamond$ (a certificate that no strategy can do better).

This mirrors the role of primal/dual solutions as (strategies)/(witnesses) throughout quantum information.

4.6 A comment on Choi-matrix formulations

The Stinespring-pair route is conceptually clean, but in implementations one often prefers an SDP that depends only on the *Choi operator*. Recall the Choi–Jamiołkowski operator of Φ : if $\{|i\rangle\}$ is an orthonormal basis of $\mathcal{X}$ and $E_{ij} = |i\rangle\langle j|$, then

$$J(\Phi) := \sum_{i,j} \Phi(E_{ij}) \otimes E_{ij} \in \mathcal{L}(\mathcal{Y} \otimes \mathcal{X}).$$

It is the unique operator satisfying

$$(\Phi \otimes \text{id}_{\mathcal{L}(\mathcal{X})})(|\Omega\rangle\langle\Omega|) = J(\Phi), \quad |\Omega\rangle = \sum_i |i\rangle|i\rangle,$$

up to the choice of basis.

For a general map Φ , one can express $|||\Phi|||_1$ as the optimum of the following SDP pair.

Primal (Choi form).

$$\begin{aligned} & \text{maximize: } \frac{1}{2} \langle J(\Phi), X \rangle + \frac{1}{2} \langle J(\Phi)^*, X^* \rangle \\ & \text{subject to: } \begin{pmatrix} \mathbb{1}_{\mathcal{Y}} \otimes \rho_0 & X \\ X^* & \mathbb{1}_{\mathcal{Y}} \otimes \rho_1 \end{pmatrix} \succeq 0, \\ & \quad \rho_0, \rho_1 \in \mathcal{D}(\mathcal{X}), \quad X \in \mathcal{L}(\mathcal{Y} \otimes \mathcal{X}). \end{aligned}$$

Dual (Choi form).

$$\begin{aligned} & \text{minimize: } \frac{1}{2} \left(\|\text{Tr}_{\mathcal{Y}}(Y_0)\|_\infty + \|\text{Tr}_{\mathcal{Y}}(Y_1)\|_\infty \right) \\ & \text{subject to: } \begin{pmatrix} Y_0 & -J(\Phi) \\ -J(\Phi)^* & Y_1 \end{pmatrix} \succeq 0, \\ & \quad Y_0, Y_1 \in \text{Pos}(\mathcal{Y} \otimes \mathcal{X}). \end{aligned}$$

Remark 4.12 (Hermiticity-preserving simplification). If Φ is Hermiticity-preserving, then $J(\Phi)$ is Hermitian and one may take $X = X^*$ and $\rho_0 = \rho_1$ in the primal. In this case the objective reduces to $\langle J(\Phi), X \rangle$ and the SDP becomes particularly transparent: it is a relaxation of maximizing $\langle J(\Phi), \cdot \rangle$ over operators bounded by $\mathbb{1}_{\mathcal{Y}} \otimes \rho$.

These Choi-based SDPs are often the most convenient starting point for numerical computation of diamond norms between explicit channels.

4.7 Complexity snippets: induced norms and contraction coefficients

It is tempting to treat “compute the operational norm” as synonymous with “solve an SDP.” The diamond norm is indeed efficiently computable from standard channel descriptions, but several *nearby* quantities become computationally difficult once one changes the stabilization or the Schatten exponents. The goal of this subsection is to give a few concrete complexity signposts.

Induced Schatten norms. For $1 \leq q, p \leq \infty$ and a linear map $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$, define the induced Schatten norm

$$\|\Phi\|_{q \rightarrow p} := \max \left\{ \|\Phi(X)\|_p : \|X\|_q \leq 1 \right\}.$$

For Hermiticity-preserving Φ one often considers the *positive-restricted* variant

$$\|\Phi\|_{q \rightarrow p}^+ := \max \left\{ \|\Phi(X)\|_p : X \succeq 0, \|X\|_q \leq 1 \right\},$$

which is the natural restriction when interpreting X as a (subnormalized) quantum state. Note that for a CPTP map Φ one has $\|\Phi\|_{1 \rightarrow 1}^+ = 1$; hardness results appear already for differences of channels or more general maps.

Hardness without complete boundedness. Recent results show that several induced norms relevant to quantum information are NP-complete to compute (in the standard promise/decision formulations), even for very restricted classes of completely positive maps (including entanglement-breaking channels):

- computing $\|\Phi\|_{1 \rightarrow 1}^+$ can be NP-complete for maps naturally presented as differences of channels, and
- for every fixed $p > 1$, computing $\|\Phi\|_{1 \rightarrow p}$ can be NP-complete even when Φ is completely positive and entanglement-breaking.

Tractability of completely bounded $1 \rightarrow p$ norms. In contrast, the *completely bounded* versions

$$\|\Phi\|_{\text{cb}, 1 \rightarrow p} := \sup_{k \geq 1} \|\Phi \otimes \text{id}_k\|_{1 \rightarrow p}$$

retain a semidefinite programming description and are polynomial-time computable for fixed p (with $p = 1$ being the diamond norm case). There are also parameter regimes in which even the *unstabilized* induced norms of CP maps become tractable; for instance, when $q \geq p$ one can often reduce $\|\Phi\|_{q \rightarrow p}$ to an efficiently computable eigenvalue-type optimization.

Contraction coefficients. A different but closely related family of quantities are *contraction coefficients* (or ergodicity coefficients), which measure worst-case decay of distinguishability under a channel. For trace distance, the contraction coefficient is

$$\eta_{\text{tr}}(\Phi) := \sup_{\rho \neq \sigma} \frac{\|\Phi(\rho - \sigma)\|_1}{\|\rho - \sigma\|_1}.$$

This coefficient controls mixing and distinguishability decay, but it also turns out to be computationally difficult: approximating $\eta_{\text{tr}}(\Phi)$ within constant factors can be NP-hard, and even deciding whether $\eta_{\text{tr}}(\Phi) = 1$ (i.e. whether the channel perfectly preserves distinguishability of *some* pair of states) can be NP-hard for natural classes of channels.

Remark 4.13 (A useful mental model). The picture that emerges is that complete boundedness often brings in additional operator-system structure that is “visible” to SDPs, while unstabilized induced norms and contraction-type quantities can encode hard combinatorics. From a practical perspective: if a quantity is defined by optimizing over *product* inputs, it is worth being cautious about computational hardness; if it is defined by optimizing over *arbitrarily entangled* inputs (a CB norm), there is a good chance an SDP exists.

5 Further reading

- J. Watrous, *The Theory of Quantum Information*, Chapter 3 (completely bounded norms) and Chapter 6 (PPT and entanglement).
- Any standard convex optimization reference for SDPs and duality (to complement Lecture 1).